



Modeling Vowel System Typology Using Iterated Confusion Minimization

John McGahay

Department of Linguistics, UCLA, USA

jmcgahay@g.ucla.edu

Abstract

This work presents a unified theory of vowel system typology based on a novel algorithm called Iterated Confusion Minimization, which simulates the interaction of optimal listeners and speakers over time. Compatible with existing sound change models that involve confusion-reducing adjustments to phonetic distributions, simulations over random initial vowel systems resemble observed sound change (e.g., chain shifts) and most often result in typologically dominant systems with minimal confusion, suggesting a diachronic explanation for the typological dominance of perceptually optimal vowel systems. In contrast to past vowel dispersion models, Iterated Confusion Minimization easily accounts for interior vowels like /ə/, allowing for prediction of the most common systems up to 11 vowels. Finally, variation in simulation output mirrors typological variation, with a tight correlation between system emergence rate and typological frequency up to 6 vowels that outperforms past baselines.

Index Terms: vowel system typology, computer modeling, dispersion theory, sound change

1. Introduction

Dispersion Theory [1, 2, 3, 4] has achieved success in modeling vowel system typology [5, 6, 7, 8, 9, 10, 11] by characterizing the most common vowel systems as perceptually optimal.

However, past dispersion models generally lack a satisfactory mechanism whereby such optimal systems emerge and come short in explaining typological variation. Although [12, 13] showed that naturalistic vowel systems can emerge at realistic rates in agent-based simulations, these simulations present an unrealistic model of sound change (although see [14, 15, 16]). Meanwhile, plausible sound change models yielding emergent dispersion [17, 18, 19] have not been computationally implemented to model vowel system typology.

To fill these gaps, the current work presents a novel algorithm called Iterated Confusion Minimization (ICM). ICM offers a realistic model of sound change based on idealized listeners and speakers, predicts realistic typological variation, and most often results in perceptually optimal, typologically dominant vowel systems when implemented over an empirically informed model of the vowel space, neatly tying together existing approaches into a unified theory of vowel system typology.

2. Iterated Confusion Minimization

The Iterated Confusion Minimization (ICM) algorithm models the interaction of mathematically optimal listeners and speakers, whose behavior is explored in the following sections.

2.1. Optimal Listener Behavior

Past work has employed Bayesian models of the optimal listener [20, 21, 22, 23]. Assuming that all phoneme categories have equal prior probability, the optimal Bayesian listener follows *maximum likelihood behavior*. Given a system of phonemes with associated phonetic distributions $S = \{\phi_1, \phi_2, \dots\}$, a maximum likelihood classifier C optimized for S categorizes a phonetic token x as follows:

$$C(x) = \underset{\phi \in S}{\operatorname{argmax}} p(x|\phi) \quad (1)$$

A maximum likelihood classifier has sharp perceptual boundaries between categories at the point where these categories are equally likely. Such perceptual boundaries for the shaded distributions in Figure 1 are marked with black vertical lines.

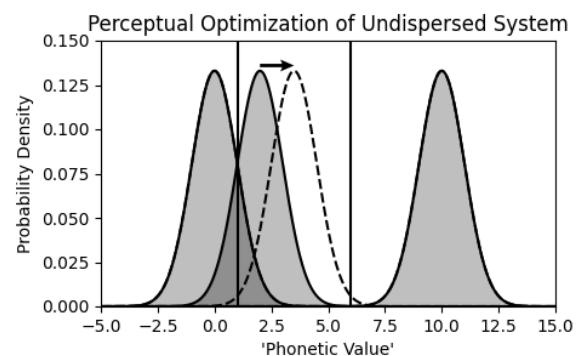
Although such perfectly sharp perceptual boundaries are an idealization of human speech perception, they allow for an intuitive definition of the confusion of a classifier C over a sound system S : the percentage of each distribution on the ‘wrong’ side of its boundaries. This is equivalent to the triangular area of distribution overlap in Figure 1, calculated as follows:

$$\begin{aligned} \operatorname{conf}_C(S) &= \frac{1}{|S|} \sum_{\phi \in S} P(C(\phi) \neq \phi) \\ &= \frac{1}{|S|} \sum_{\phi \in S} \int p(x|\phi) P(C(x) \neq \phi) dx \end{aligned} \quad (2)$$

2.2. Optimal Speaker Behavior

Although one might naïvely expect that the optimal speaker will minimize a listener’s confusion by converging to the exact phonetic system of that listener, Figure 1 shows that a speaker can

Figure 1: *Illustration of the confusion paradox. Adopting a sound system different from that of a listener reduces confusion.*



in fact do better. By shifting the middle distribution with a solid outline to the dashed distribution, a speaker can reduce the percentage of that distribution on the wrong side of its perceptual boundaries. This will reduce the listener's confusion, *even if the listener is expecting to hear the original set of distributions*, in effect ‘gaming’ that listener's perception.

We will call this observation the *confusion paradox*. Given a set of phonetic distributions S and a classifier C optimized over S , the optimal set of phonetic distributions S' that minimizes the confusion rate of C is not in general equivalent to the original set of phonetic distributions S for which C is optimized. In particular, given a set of possible phonetic distributions Φ and a listener C optimized over $S \subseteq \Phi$, an optimal speaker should adopt the system $S' \subseteq \Phi$:

$$S' = \left\{ \underset{\phi' \in \Phi}{\operatorname{argmin}} P(C(\phi') \neq \phi) \mid \phi \in S \right\} \quad (3)$$

$$= \left\{ \underset{\phi' \in \Phi}{\operatorname{argmin}} \int p(x|\phi') P(C(x) \neq \phi) dx \mid \phi \in S \right\}$$

2.3. The Iterated Confusion Minimization Algorithm

Once an optimal speaker adjusts their phonetic distributions to minimize an optimal listener's confusion, that listener should adjust their perception to accommodate the speaker's new distributions. However, that listener's updated perceptual boundaries are susceptible to the same confusion paradox that motivated the speaker to shift distributions in the first place. Thus, we expect the optimal speaker and listener to iteratively adjust their perception and production in response to one another until some perceptually self-optimal equilibrium system is reached.

This interaction of optimal listener and speaker over time is modeled using Iterated Confusion Minimization (ICM), given in Algorithm 1. Figure 2 illustrates how ICM can model realistic sound change like chain shifts, tending to produce naturalistic vowel systems with high dispersion and low confusion when implemented over the vowel space, as detailed in Section 3.

ICM shares its basic mechanism, i.e. restructuring of phonetic distributions to include a smaller proportion of easily confused tokens, with [17]'s account of chain shifts within Exemplar Theory [24] and [18]'s account of emergent sibilant dispersion within Stochastic Optimality Theory [25, 26]. By abstracting away from phonological representations to a strictly computational-level analysis [27] of human phonetic behavior, ICM renders the relationship between this mechanism and listener confusion mathematically explicit.

Algorithm 1 Iterated Confusion Minimization (ICM)

Input: $S_0 \subseteq \Phi$: an initial sound system S_0 drawn from Φ , where Φ is the set of possible phonetic distributions

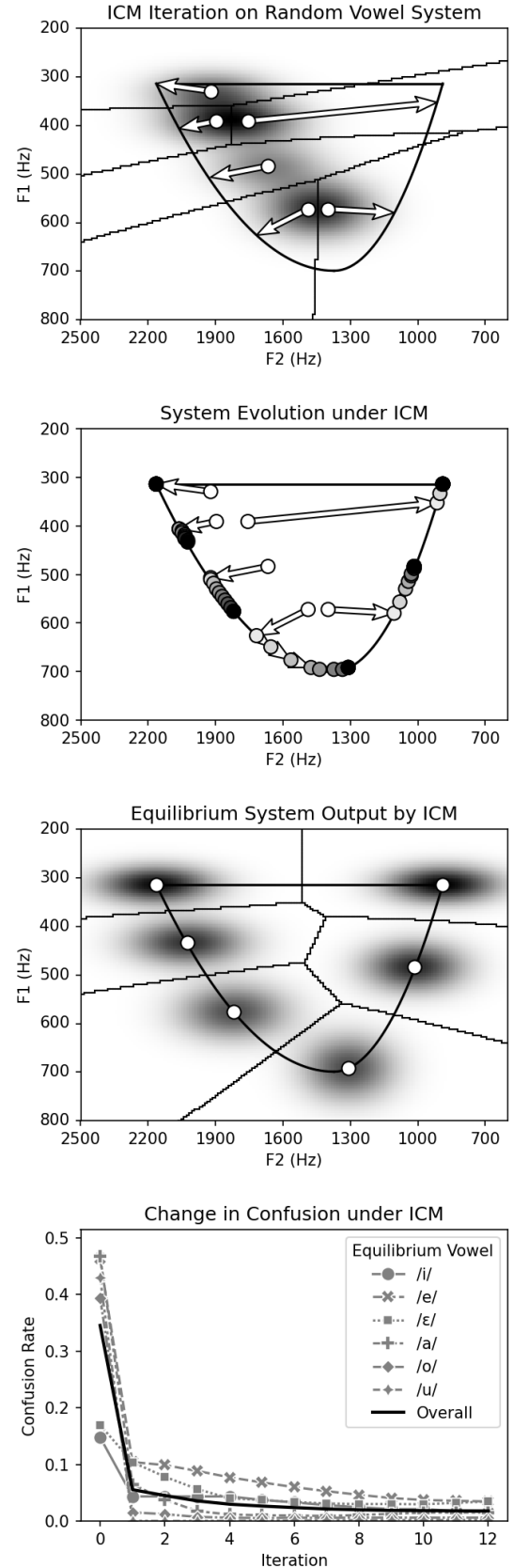
Output: $S_i \subseteq \Phi$: the sound system S_i after i iterations

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1:  $i \leftarrow 0$ 
2: repeat
3:   ▷ Optimal listener behavior
4:    $C_i(x) \leftarrow \operatorname{argmax}_{\phi \in S_i} p(x|\phi)$ 
5:   ▷ Optimal speaker behavior
6:    $S_{i+1} \leftarrow \{\operatorname{argmin}_{\phi' \in \Phi} P(C_i(\phi') \neq \phi) \mid \phi \in S_i\}$ 
7:    $i \leftarrow i + 1$ 
8: until  $S_i = S_{i-1}$ 
9: return  $S_i$ 

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Figure 2: ICM simulation over a random initial 6-vowel system demonstrating a chain shift along the vowel space periphery.



3. Implementing ICM over the Vowel Space

ICM over the vowel space requires a mathematically well-defined model of possible vowel distributions, provided in the next sections followed by example simulation results. Attempts were made in selection of this model to balance simplicity, comparability with past work, and empirical groundedness.

3.1. Defining the Vowel Space

We will consider a two-dimensional space whose vowels have F1 and F2 means (μ_1, μ_2) in Hz limited by the following equations, adapted from [1] to match the vowel space reported by [11] for /ieaou/, the world's most common vowel system:

$$314 \leq \mu_1 \leq 699 \quad (4)$$

$$\mu_2 \leq 1373 + (2164 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}}$$

$$\mu_2 \geq 1373 + (889 - 1373) \sqrt{\frac{699 - \mu_1}{699 - 314}}$$

Because confusion depends on not distance but overlap, which does not vary across scales, we are not required to use the psychoacoustic scales of past dispersion models [1, 3]. Instead, we use Hz since it is the most basic, widely used frequency scale.

3.2. Modeling Phonetic Distributions of Vowels

Vowels are modeled as bivariate Gaussians with 0 covariance:

$$(F1, F2) \sim \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \right) \quad (5)$$

where F1 and F2 standard deviations (σ_1, σ_2) are a function of formant means based on linear regression over z-scored data for 19 speakers of 21 languages from 10 studies [28, 29, 30, 31, 32, 33, 34, 35, 36, 37] with random effects for speaker and study:

$$z_1 = \frac{\mu_1 - 498.7212}{163.5568} \quad z_2 = \frac{\mu_2 - 1460.05}{543.4261} \quad (6)$$

$$\sigma_1 = 34.9728 + 8.0738z_1 - 0.2633z_2 - 0.3906z_1z_2$$

$$\sigma_2 = \lambda(86.803 - 6.611z_1 + 6.741z_2 + 5.136z_1z_2)$$

Following past work [3], this model also experiments with F2 scaling, implemented as a weight $1.0 \leq \lambda \leq 4.0$ on σ_2 .

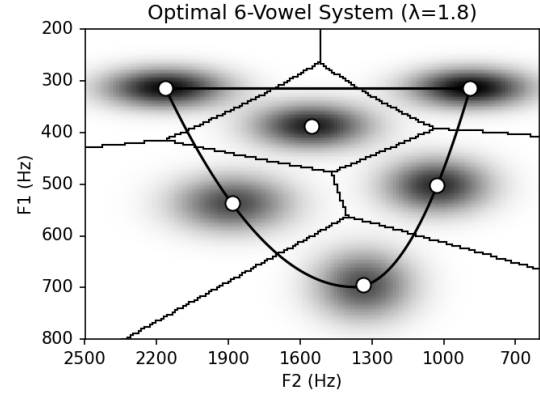
3.3. Example Result: Predicting Chain Shifts

Figure 2 shows an example simulation over a random initial 6-vowel system using F2 weight $\lambda = 1.8$. After an initial iteration over an undispersed system (first plot), system evolution follows an incremental chain shift along the vowel space periphery (second plot), eventually reaching a perceptually self-optimal system (third plot). Each iteration results in a reduction in the overall confusion rate (last plot). Given the ubiquity of such chain shifts in the phonological history of the world's languages [38] and the result of a typologically common vowel system /ieaou/ [10], this simulation shows that ICM provides an appealing model of emergent dispersion through sound change.

3.4. Example Result: Predicting Interior Vowels

Although the system /ieaou/ of Figure 2 is attested, the typologically dominant system is actually /ieaouə/ [10, 11]. This system, shown in Figure 3, is also stable under ICM and in fact has the lowest confusion among 6-vowel ICM outputs. Beyond

Figure 3: The optimal 6-vowel system has an interior vowel.



the agreement between typological dominance and perceptual optimality, this result is attractive in its narrower acoustic structure. The predicted high-mid central vowel [ə] closely matches the acoustic typology of the dominant 6-vowel system [11], and contrasts with the systematic failure of past dispersion models [3] to predict vowels like /ə/ in the vowel space interior.

4. Generating Vowel System Typology

Since multiple systems are stable under ICM, the next sections evaluate the ICM emergence rate of different vowel systems as a model of the typological frequencies of those systems. First, however, we require a typological baseline for this comparison.

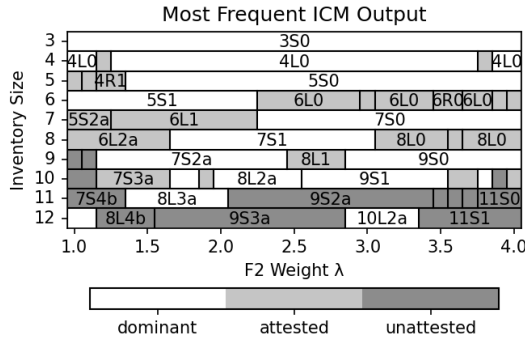
4.1. Baseline Typological Frequencies of Vowel Systems

Our typological baseline derives from [10]'s analysis of UPSID [9], a genetically and areally balanced survey of the world's sound systems. However, since [10] make several narrow distinctions between systems that [11]'s acoustic typology suggests are artifacts of transcription choice, [10]'s analysis has been recast into the taxonomy of [11]. [11] assigns each system a label $\#_p X \#_{np}$, where $\#_p$ is the number of peripheral vowels in the system, X takes the value L, R, or S for left-crowded, right-crowded, or symmetric inventories, and $\#_{np}$ is the number of non-peripheral vowels. Additionally, inventories with two or more non-peripheral vowels are given the suffix 'a' if these are arranged vertically and 'b' if they include a horizontal contrast. Table 1 lists the dominant systems according to this baseline.

Table 1: Typologically dominant vowel systems in UPSID [9].

Inventory Size	System	Example IPA	Frequency
3 (n=19)	3S0	/iau/	1.000
4 (n=25)	4L0	/ieau/	0.560
5 (n=109)	5S0	/ieaou/	0.890
6 (n=60)	5S1	/ieaouə/	0.683
7 (n=44)	7S0	/ieəaou/	0.523
8 (n=19)	7S1	/ieəaouə/	0.421
9 (n=24)	7S2a	/ieəaouia/	0.292
	9S0	/ieəaouu/	0.292
10 (n=8)	8L2a	/ieəaouia/	0.250
	9S1	/ieəaouuə/	0.250
11 (n=4)	8L3a	/ieəaouyθə/	0.667

Figure 4: Vowel systems most frequently output by ICM across parameter settings, shaded by attestation in UPSID [9].



4.2. Predicting Dominant Vowel Systems

Figure 4 shows the most frequent vowel systems output by ICM across 310 parameter settings (31 values of F2 weight λ and 10 inventory sizes from 3 to 12) based on 100 randomly initialized simulations per parameter setting (31,000 total simulations).

We can see that every typologically dominant system is output by some parameter setting and a single vowel space model ($\lambda = 1.8$, among others) can simultaneously predict a dominant vowel system for every inventory size except for 7 up to 11 vowels. This is a marked improvement over the baseline set by [3], who only report successful predictions up to 7 vowels and are likewise unable to simultaneously predict the dominant 6 and 7 vowel systems using the same F2 weight. This improvement appears due to the ability of ICM to predict interior vowels common in larger vowel inventories, e.g. 7S2a /ieɛəɔuɪə/.

4.3. Predicting Typological Frequencies

Figure 5 shows a strong correlation of ICM emergence rate with typological frequency up to 7 vowels, covering 81% of the primary vowel systems in [10]’s analysis of UPSID [9]. An appropriate quantitative measure for evaluating this correspondence is Kullback-Leibler (KL) divergence [39] of typological frequencies P from emergence rates Q of each vowel system S :

$$D_{KL}(P||Q) = \sum_S P(S) \log \left(\frac{P(S)}{Q(S)} \right) \quad (7)$$

Figure 5: ICM emergence vs. typology up to 7 vowels.

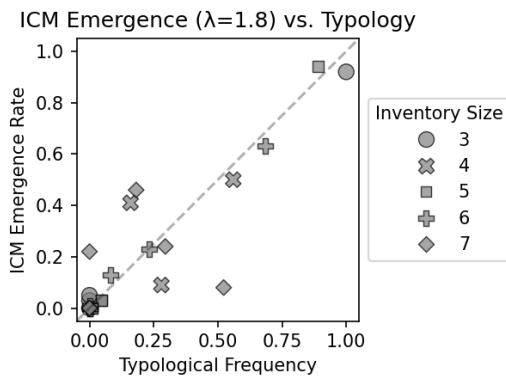


Table 2: KL divergence of typology from emergence rate.

Inventory Size	KL Divergence	
	ICM ($\lambda = 1.8$)	de Boer (1999) [12]
3	0.084	0.248
4	0.231	∞
5	0.009	0.009
6	0.022	0.028
7	0.874	0.383

Intuitively, KL divergence provides a measure of how ‘surprising’ typological frequencies are when emergence rates are used as a model. A lower value indicates a better model. Table 2 shows that the best ICM model ($\lambda = 1.8$) meets or outperforms the baseline set by [12] in predicting typological frequencies of systems up to 6 vowels.

5. Discussion

The current work builds on the past success of Dispersion Theory in characterizing the most common vowel systems across the world’s languages as perceptually optimal. Adopting a measure of perceptual contrast directly rooted in a mathematical model of speech perception allows for extension of this approach to larger inventory sizes including interior vowels like /ə/, which are poorly explained by past dispersion models. Furthermore, by modeling not only which systems are optimal but also how such systems might emerge through sound change, the current work attains greater explanatory power in accounting for typological variation between vowel systems.

Although this work shares an emphasis on the optimality of sound systems with the synchronic framework of Optimality Theory [40], which has underpinned more recent work within Dispersion Theory [4, 41, 42], ICM differs from Optimality Theory in attributing an aspect of phonological typology to a mechanism of sound change rather than the innate structure of individuals’ phonological grammar. This alternate view of typology as emergent from sound change is shared with the diachronic framework of Evolutionary Phonology [17], which however disputes the optimality of phonological systems. The current work thus suggests a compromise between these two perspectives: phonological systems tend to be optimal, but this optimality can emerge from sound change rather than needing to be explicitly encoded in the synchronic grammar.

Given the assertion of Evolutionary Phonology [17] that sound change is not goal-directed, it might seem contradictory that perceptually optimal systems are shown to easily arise through a sound change mechanism broadly compatible with [17]’s own account of chain shifts. This discrepancy can be resolved through an understanding of this sound change mechanism as a form of local optimization. Local optimization algorithms work by breaking down the computationally difficult problem of global optimization into many incremental, short-sighted steps. Similarly, the current work shows that incremental sound change can produce sound systems that are globally optimal with respect to perception even though this is not a goal explicitly encoded into the sound change mechanism. When individuals make short-sighted phonetic adjustments at the level of individual speech sounds in an attempt to act as effective listeners and speakers of their language, these adjustments can accumulate over time into optimal structure at the level of entire sound systems.

6. Acknowledgments

I would like to thank Bruce Hayes, Megha Sundara, Meg Cychosz, Patricia Keating, Roy Becker-Kristal, the UCLA phonetics seminar, and three anonymous reviewers for their thoughtful comments on this work. I would also like to thank research assistants Valerie So, Shawn Grant, Remi Akopians, Bill Tran, Makenna Kumlue, and Christopher Yang for their help creating the vowel acoustics data set used to fit the formant standard deviation model given in Section 3.2. This research was funded in part by the UCLA Division of Graduate Education through its Graduate Summer Research Mentorship program.

7. References

- [1] J. Liljencrants and B. Lindblom, "Numerical simulation of vowel quality systems: The role of perceptual contrast," *Language*, pp. 839–862, 1972.
- [2] B. Lindblom, "Phonetic universals in vowel systems," *Experimental phonology*, pp. 13–44, 1986.
- [3] J.-L. Schwartz, L.-J. Boë, N. Vallée, and C. Abry, "The dispersion-focalization theory of vowel systems," *Journal of phonetics*, vol. 25, no. 3, pp. 255–286, 1997b.
- [4] E. Flemming, "Auditory representations in phonology," Ph.D. dissertation, Ph. D. Dissertation, University of California, Los Angeles, 1995.
- [5] N. S. Trubetzkoy, *Grundzüge der Phonologie*. Van den Hoeck & Ruprecht, 1939.
- [6] C. F. Hockett, *A Manual of Phonology*. Baltimore: Waverly Press, 1955.
- [7] P. Sedlak, "Typological considerations of vowel quality systems," *Working Papers on Language Universals*, vol. 1, pp. 1–40, 1969.
- [8] J. Crothers, "Typology and universals of vowel systems in phonology," *Universals of human language*, vol. 2, pp. 95–152, 1978.
- [9] I. Maddieson, *Pattern of Sounds*. Cambridge, UK: Cambridge University Press, 1984.
- [10] J.-L. Schwartz, L.-J. Boë, N. Vallée, and C. Abry, "Major trends in vowel system inventories," *Journal of phonetics*, vol. 25, no. 3, pp. 233–253, 1997a.
- [11] R. Becker-Kristal, *Acoustic typology of vowel inventories and Dispersion Theory: Insights from a large cross-linguistic corpus*. University of California, Los Angeles, 2010.
- [12] B. de Boer, "Self organization in vowel systems," Ph.D. dissertation, Ph. D. Dissertation, AI-Laboratory Vrije Universiteit Brussel, 1999.
- [13] —, "Self-organization in vowel systems," *Journal of phonetics*, vol. 28, no. 4, pp. 441–465, 2000.
- [14] J. N. Stanford and L. A. Kenny, "Revisiting transmission and diffusion: An agent-based model of vowel chain shifts across large communities," *Language variation and change*, vol. 25, no. 2, pp. 119–153, 2013.
- [15] M. Sósuthy, "Understanding change through stability: A computational study of sound change actuation," *Lingua*, vol. 163, pp. 40–60, 2015.
- [16] J. Harrington, F. Kleber, U. Reubold, F. Schiel, and M. Stevens, "Linking cognitive and social aspects of sound change using agent-based modeling," *Topics in cognitive science*, vol. 10, no. 4, pp. 707–728, 2018.
- [17] J. Blevins, *Evolutionary phonology: The emergence of sound patterns*. Cambridge University Press, 2004.
- [18] P. Boersma and S. Hamann, "The evolution of auditory dispersion in bidirectional constraint grammars," *Phonology*, vol. 25, no. 2, pp. 217–270, 2008.
- [19] B. Vaux and B. Samuels, "Explaining vowel systems: dispersion theory vs natural selection," *The Linguistic Review*, vol. 32, no. 3, pp. 573–599, 2015.
- [20] M. Clayards, M. K. Tanenhaus, R. N. Aslin, and R. A. Jacobs, "Perception of speech reflects optimal use of probabilistic speech cues," *Cognition*, vol. 108, no. 3, pp. 804–809, 2008.
- [21] D. Norris and J. M. McQueen, "Shortlist b: a bayesian model of continuous speech recognition," *Psychological review*, vol. 115, no. 2, p. 357, 2008.
- [22] N. H. Feldman, T. L. Griffiths, and J. L. Morgan, "The influence of categories on perception: explaining the perceptual magnet effect as optimal statistical inference," *Psychological review*, vol. 116, no. 4, p. 752, 2009.
- [23] D. F. Kleinschmidt, "Structure in talker variability: How much is there and how much can it help?" *Language, cognition and neuroscience*, vol. 34, no. 1, pp. 43–68, 2019.
- [24] J. Pierrehumbert, "Exemplar dynamics: Word frequency, lenition, and contrast," *Frequency and the Emergence of Linguistic Structure/John Benjamins*, 2001.
- [25] P. Boersma *et al.*, "How we learn variation, optionality, and probability," in *Proceedings of the Institute of Phonetic Sciences of the University of Amsterdam*, vol. 21. Amsterdam, 1997, pp. 43–58.
- [26] P. Boersma and B. Hayes, "Empirical tests of the gradual learning algorithm," *Linguistic inquiry*, vol. 32, no. 1, pp. 45–86, 2001.
- [27] D. Marr, *Vision*. W.H. Freeman, 1982.
- [28] P. Holtse, "Spectrographic analysis of english vowels," *Annual Report of the Institute of Phonetics University of Copenhagen*, vol. 6, pp. 1–48, 1972.
- [29] T. L. Thongkum, "An acoustic study of the register complex in kui (suai)," *Mon-Khmer Studies*, vol. 15, pp. 1–19, 1986.
- [30] M. S. McCord, *Acoustical and autosegmental analyses of the Mayo vowel system*. The University of Texas at Arlington, 1989.
- [31] J.-O. Svantesson, "Phonetic evidence for the great mongolian vowel shift," in *Proceedings of the XIIIth International Congress of Phonetic Sciences (ICPhS 95)*, vol. 1, 1995, pp. 416–419.
- [32] K. S. Olson, "The phonology and morphology of mono," Ph.D. dissertation, UMI Ann Arbor, 2001.
- [33] S. Calamai, P. Sorianoello *et al.*, "Aspetti stilistici del vocalismo romano," *Quaderni del Laboratorio di Linguistica della Scuola Normale Superiore di Pisa*, vol. 4, no. 2003, pp. 27–41, 2003.
- [34] S. Calamai, "The pisan vowel system of read and semispontaneous speech. an exploratory contribution," 2004.
- [35] C. Thornell and Y. Nagano-Madsen, "Preliminaries to the phonetic structure of the bantu language mpiemo," *Africa & Asia: Göteborg working papers on Asian and African languages and literatures*, vol. 4, pp. 163–180, 2004.
- [36] W.-y. Chiang and F.-m. Chiang, "Vowel dispersion in truku," *Concentric: Studies in Linguistics*, vol. 31, no. 1, pp. 21–45, 2005.
- [37] R. Mayr and H. Davies, "The monophthongs and diphthongs of north-eastern welsh: An acoustic study," in *Tenth Annual Conference of the International Speech Communication Association*, 2009.
- [38] W. Labov, *Principles of Linguistic Change, Volume 1: Internal Factors*. Blackwell, 1994, vol. 1.
- [39] S. Kullback and R. A. Leibler, "On information and sufficiency," *The annals of mathematical statistics*, vol. 22, no. 1, pp. 79–86, 1951.
- [40] A. Prince and P. Smolensky, "Optimality theory: Constraint interaction in generative grammar," *University, New Brunswick, and University of Colorado*, 1993.
- [41] E. Flemming, "Evidence for constraints on contrast: The dispersion theory of contrast," *UCLA working papers in Phonology*, vol. 1, pp. 86–106, 1996.
- [42] —, "Contrast and perceptual distinctiveness," *Phonetically based phonology*, vol. 232, p. 276, 2004.